UF Herbert Wertheim College of Engineering
UNIVERSITY of FLORIDA

Lecture 5C – Auditory & Somatosensory Systems

EIN 3241: Human Factors and Ergonomics
Spring 2026

February 25, 2026

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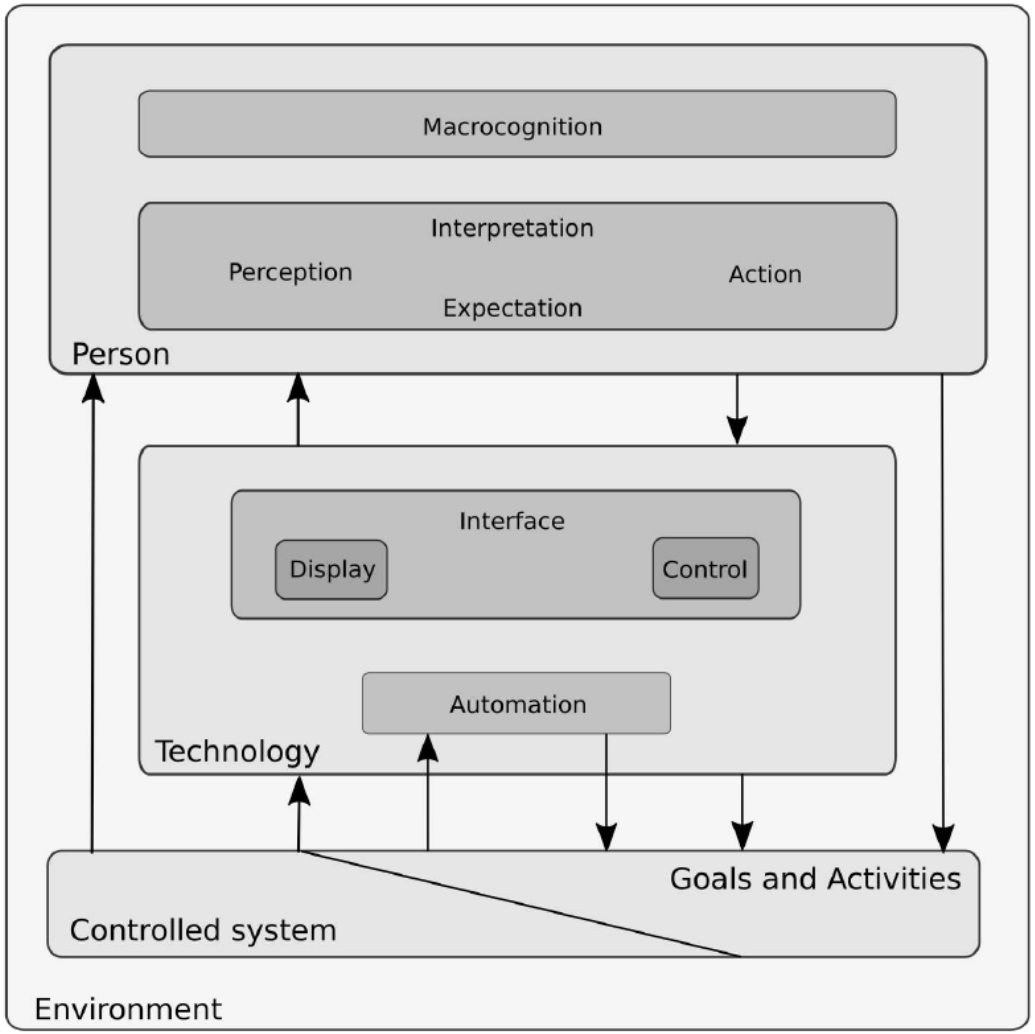
Admin Issues

- Quiz on Friday
- Midterm
  - Monday, March 9
  - 8:20 PM – 10:10 PM in FLG270
  - Up until Cognition and Human Information Processing

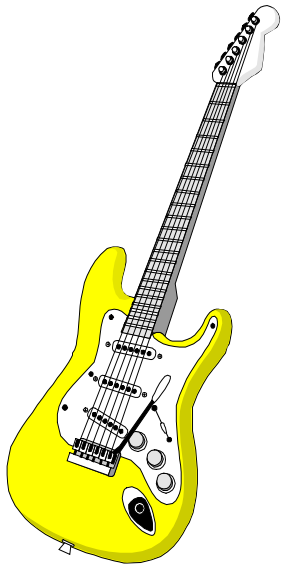
Again… Why are we talking about this

- Sensation and perception are the first stages of the human cognitive process
- Human perception is often non-linear
- The systems view of human-interactive systems.
- About 95-99% of information is channeled through the eyes or ears…

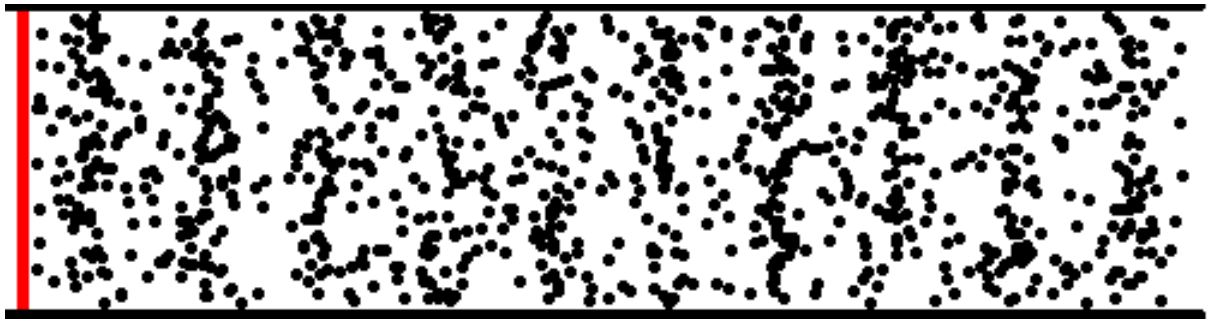


Auditory stimulus

- Sound created by vibrations (compressions and refractions) from some source:



Air molecules



1) An object makes a sound by vibrating

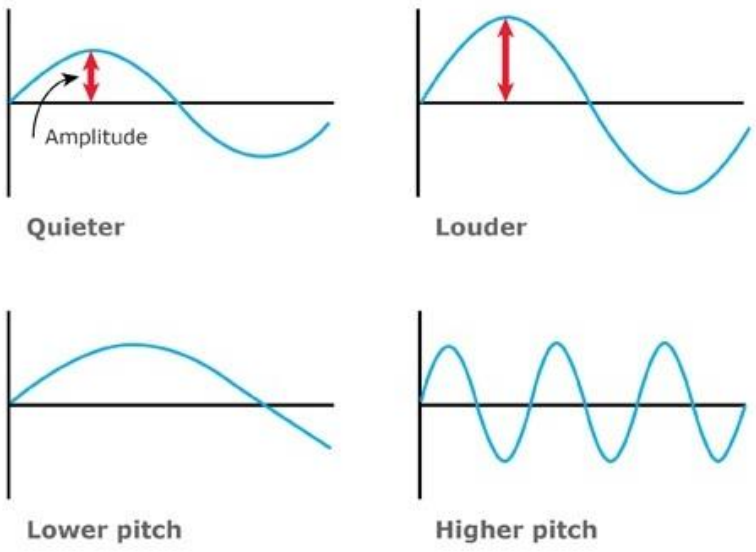
2) The vibrations pass through air by making air molecules vibrate

3) These vibrations are picked up by the ear

- Sound pressure - amount of pressure variation relative to atmospheric pressure caused by sound

Physical properties of sound

- Sound waves have two primary attributes…
  - Amplitude (in dB)
    - Amount of energy in sound wave
    - Humans perceive amplitude as loudness
  - Frequency (in Hz)
    - Humans hear frequencies of sound as different pitches
      - Low frequency sound has low pitch (e.g., foghorn on ship).
      - High-frequency sound has high pitch (e.g., whistle or siren).

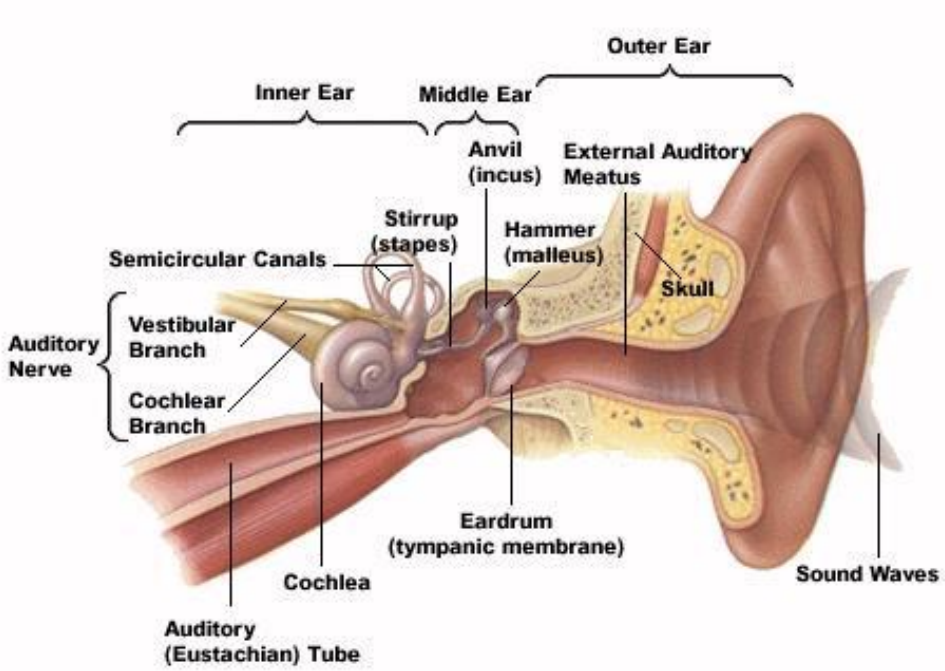


500 Hz

1000 Hz

Anatomy of the ear

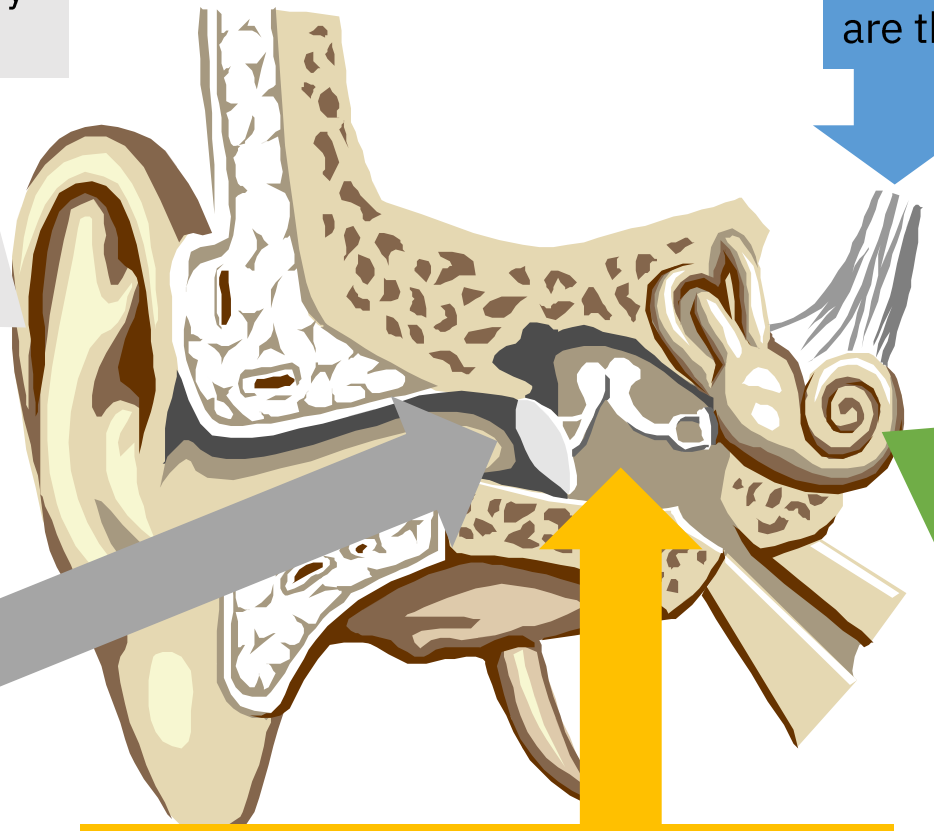
- Outer ear:
  - Pinna/Auricle (fleshy portion) and ear canal
  - Modification of incoming pressure wave for localization
  - Modification of pressure wave/acoustic wave
- Middle ear:
  - Eardrum and ossicles (malleus, incus, and stapes)
  - Conversion from airborne vibration to structural vibration
  - Impedance matching of air vibration to fluid motion (in inner ear)
- Inner ear:
  - Semi-circular canals, cochlear duct and organ of corti
  - Transforms mechanical vibration to nerve impulse.
  - Hair cells (cilia) move with fluid motion and generate neural signals
    - Nerves at beginning of channel (base) respond to short wavelengths; high pitch sounds
    - Nerves at end of channel (apex) respond to long wavelengths; low pitch sounds



How does the ear work?

1) Sound waves are "funnelled" into the ear by the pinna

5) The electrical signals are then sent to the brain



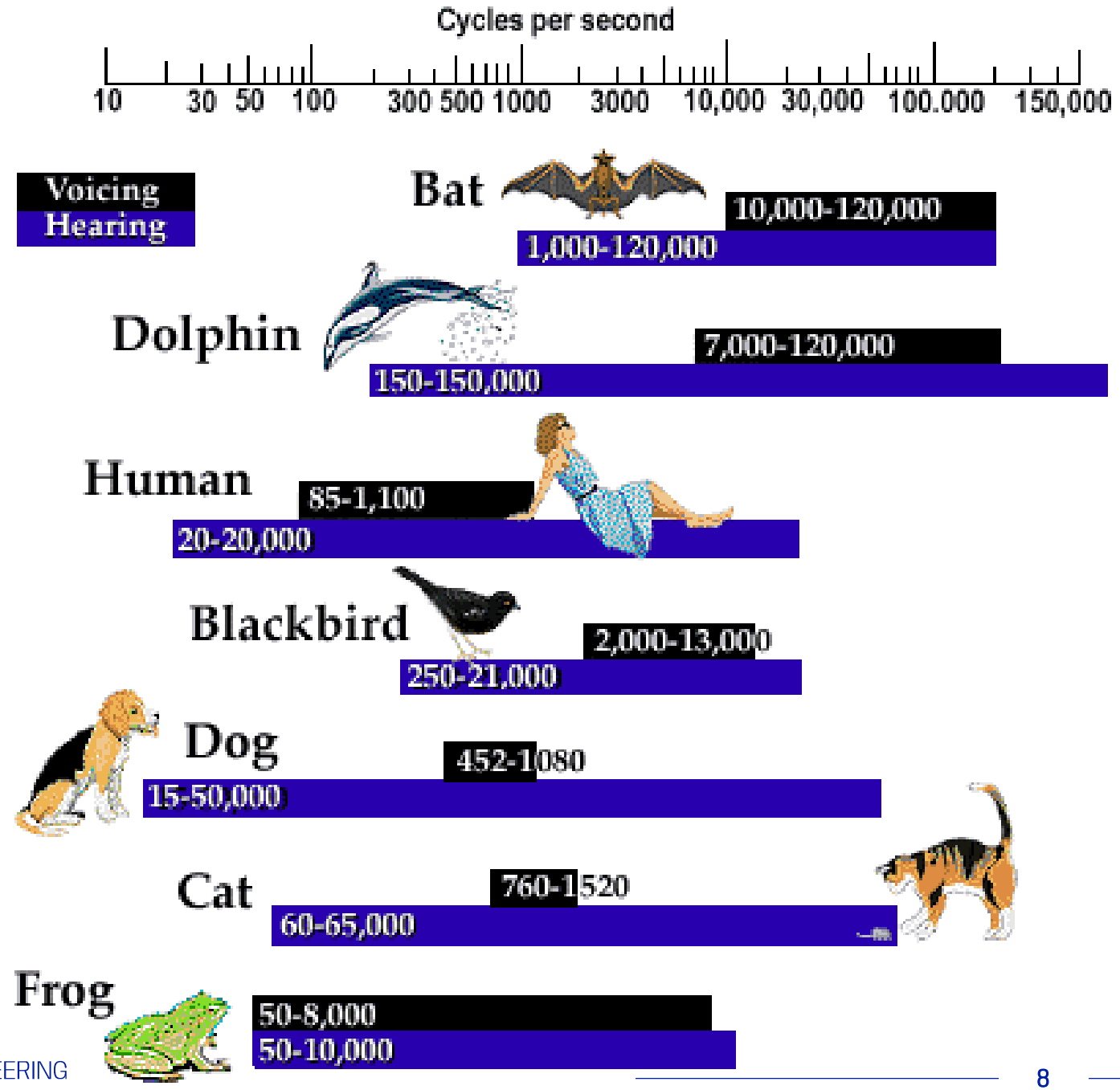
4) These vibrations are turned into electrical signals in the cochlea

2) These vibrations make the ear drum vibrate

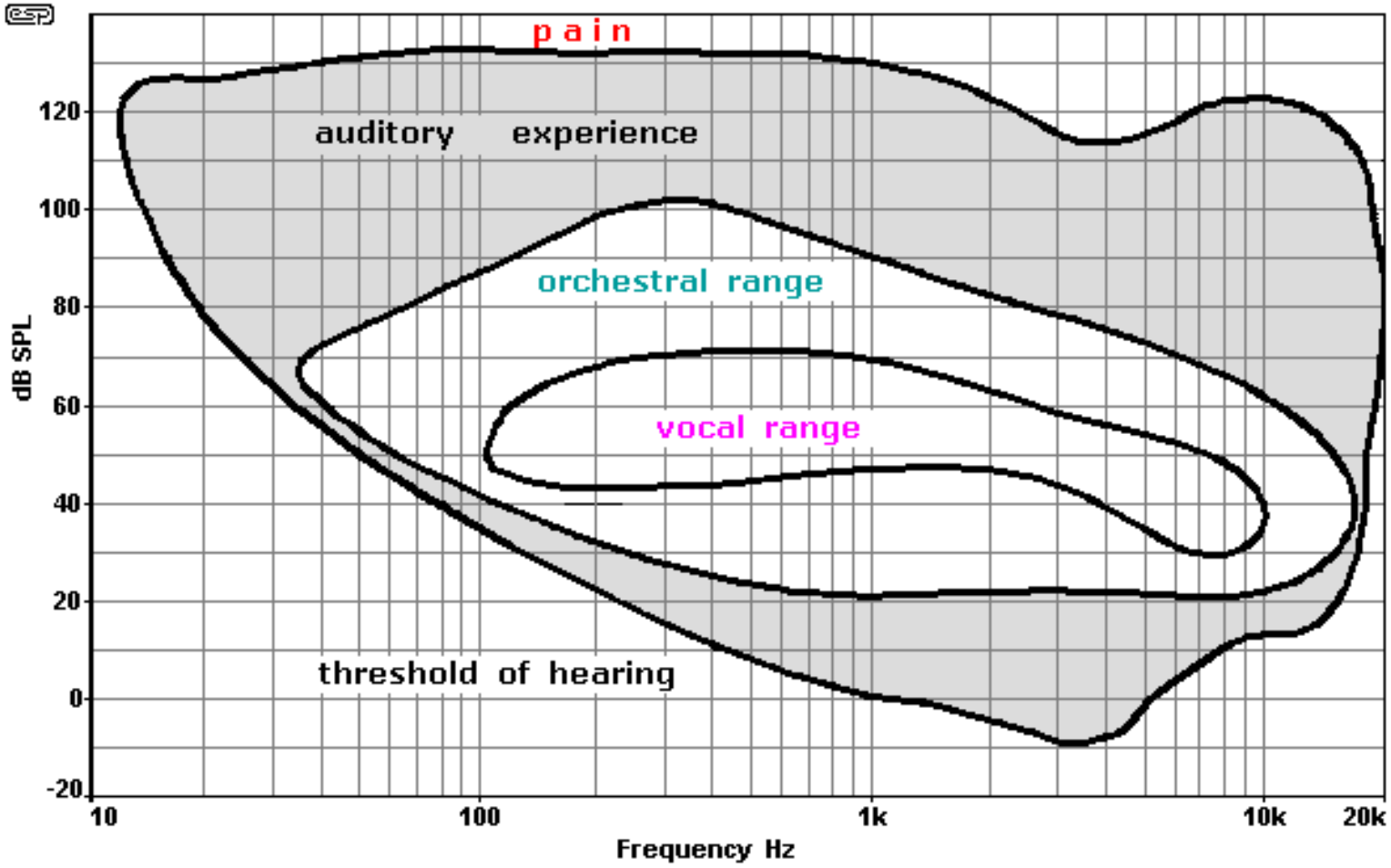
3) These vibrations make the ear bones vibrate

Frequency Ranges of Voicing and Hearing

- Humans:
  - 20 to 20,000 Hz [cycles/second]
  - 1000 to 4000 Hz most sensitive
  - Average person can distinguish 2000 pitches (musicians many more)

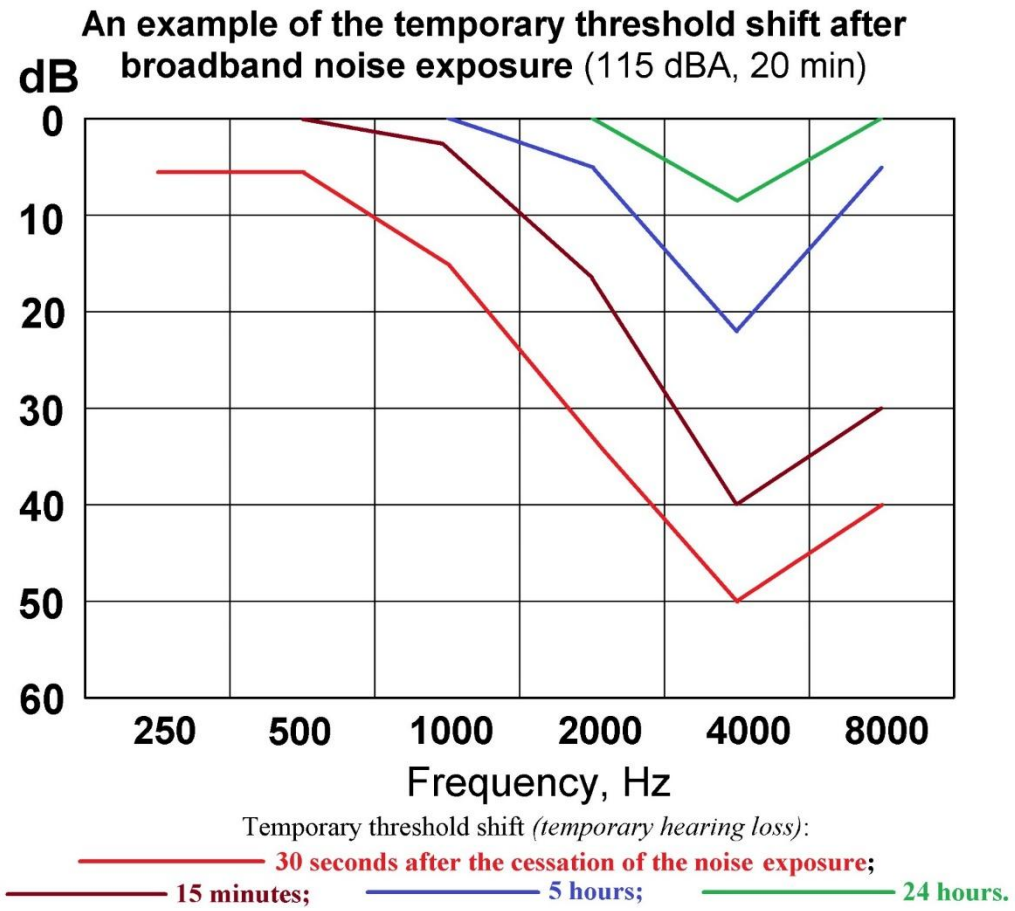


Amplitude and Frequency



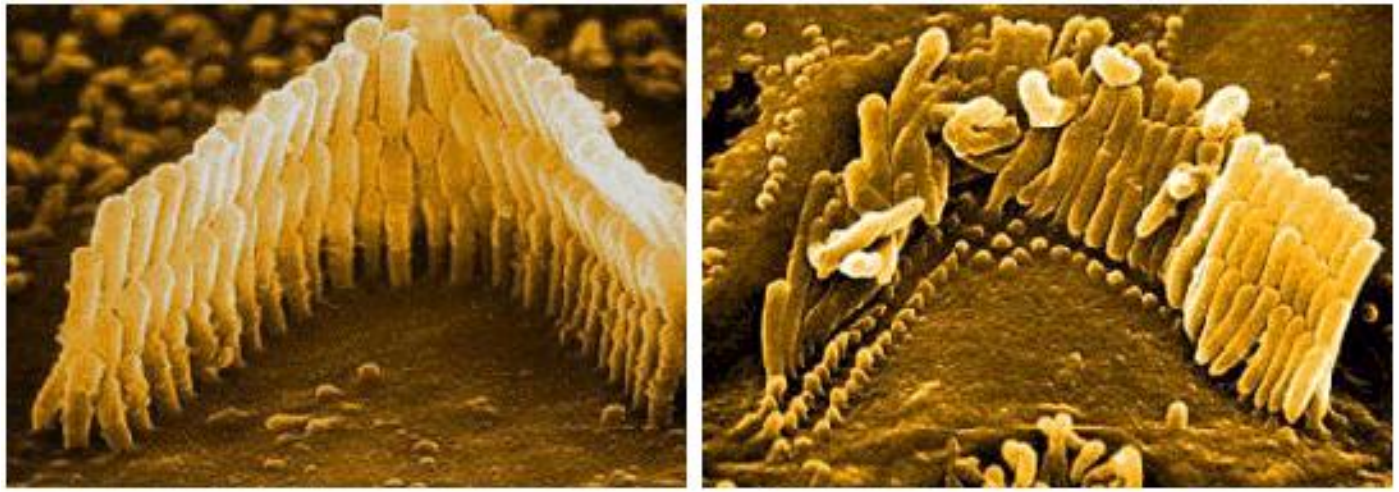
Hearing Loss

- Masking:
  - Perception of one sound affected by presence of another sound
  - Alarms should be designed 15 dB above ambient noise levels (general "rule of thumb") to ensure perception
- Temporary Threshold Shift (TTS):
  - Temporary loss of sensitivity due to acute exposure to high amplitude signals
  - e.g., "carryover" effect from using loud machines when trying to do something quieter
  - e.g., leaving a loud concert and trying to have conversation
- Permanent Threshold Shift (PTS):
  - Presbycusis – age-related hearing loss (primarily in high frequency)
  - Sensorial-neural loss – hearing loss in high frequency range (damage to hair cells)
  - Conductive loss – hearing loss due to earwax, ossification of bones of middle ear, or trauma (i.e., ruptured eardrum)



Nature of auditory impairment (PTS)

- Hair cell damage when exposed to loud music or noise
- Loud sounds = high amplitude pressure waves (e.g., standing next to subwoofer and "feeling" bass response)
- High pressure waves bend cilia too far; sometimes to point of damage.
- Hair cells do NOT grow back and leads to permanent loss



Normal, healthy outer hair cells (L) vs. damaged outer hair cells (R)

Combining sources and decibel calculation

- How do you calculate combined noise exposure when a person is exposed to more than one source simultaneously?

$$SPL_{Tot} = 10 \log_{10} \left(\sum_{i=1}^{n} 10^{\left(\frac{SPL_i}{10}\right)}\right)$$

- Formula requires division of dB measures by 10 before calculating power. Necessary for unit conversion.
- E.g., What is the combined (noise) level of two sounds of 86 and 96 decibels?

$$= 10 \log_{10} (10^{86/10} + 10^{96/10})$$

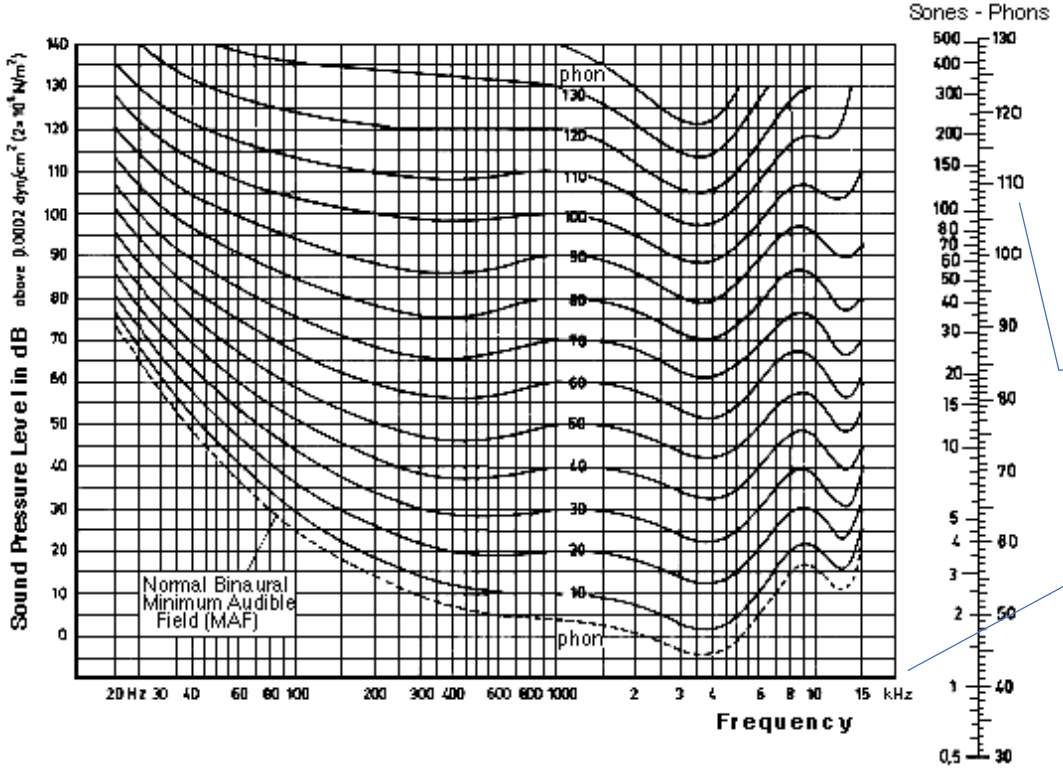
$$= 96.4 \text{ dB}$$

Psychophysical scaling of sound

- Perceived loudness is correlated with sound intensity:
  - Perception is affected by both intensity and pitch
  - Two sounds of equal intensity (e.g., 50 dB) but different in pitch (e.g., 500 Hz vs. 1000 Hz) are not of equal "loudness"
- Sone:
  - Scale relating perceived / psychological experience of loudness to physical intensity
  - 1 sone = loudness of sound at 40 dB and 1000 Hz (arbitrarily selected)
  - Perceived loudness (sone) doubles with ~10 dB increase in intensity (holding pitch fixed)
  - A **phon** is a unit of *perceived loudness*, tied to the decibel scale at **1000 Hz**

Equal loudness contours

- Human hearing sensitivity varies with frequency:
  - Most sensitive to 2-4 kHz range of stimuli.
  - Equal Loudness Curves (ELCs) have been constructed to show variation in perceived intensity with frequency for human ear.
  - Sound perceived as loud as 40 dB sound at 1,000 Hz is said to have loudness = 1 sone = 40 phons



The **scale on the right** labeled "Sones – Phons" just gives a cross-reference to perceived loudness in *sones*. E.g., 40 phons = 1 sone

Intensity

Perceived loudness

Frequency

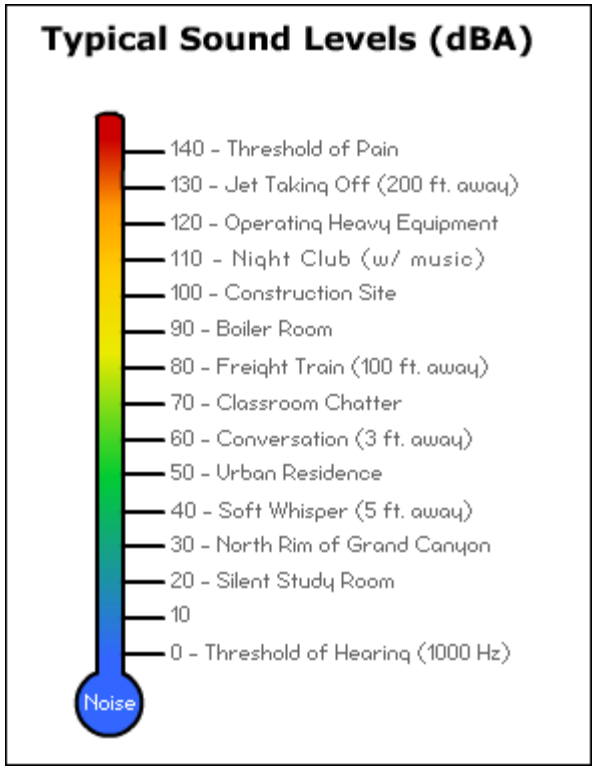
Phon calculations

- At 1000 Hz phons are approx. equal to sound intensity
- If the sound of a dog barking at 80 dB SPL is judged to be as loud as a 1,000Hz tone at 90 dB SPL, the perceived loudness of the dog's barking is _____?

a) 80 phons
b) 80 dB
c) 80 sones
d) 90 phons
e) 90 dB
f) 170 phons
g) 1,000 Hz

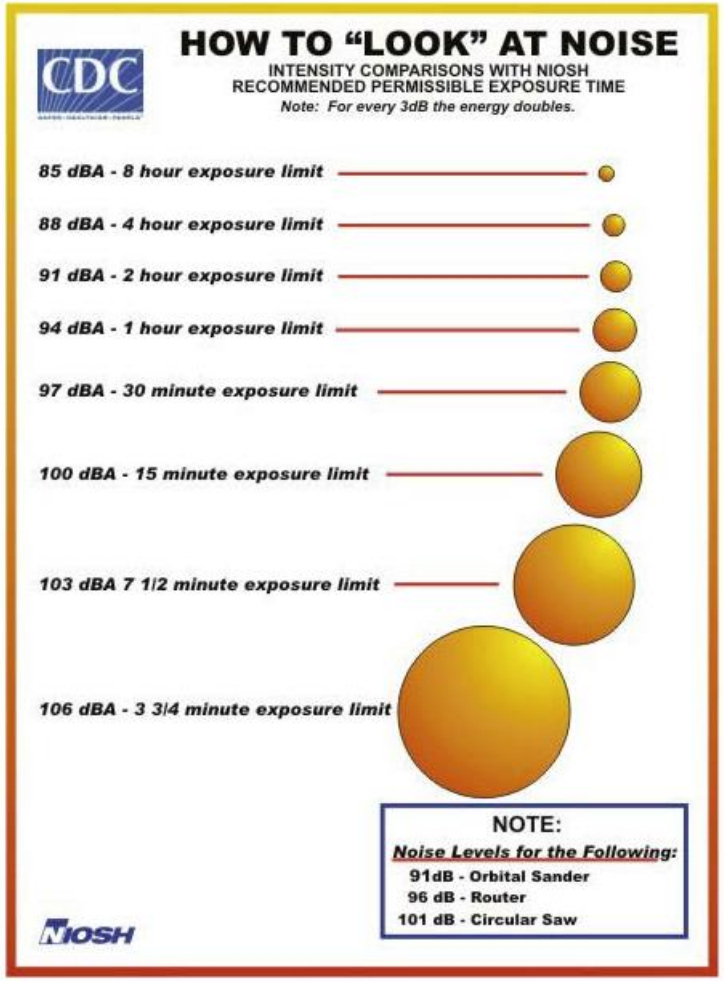
Human factors application

- Intense sound and noise exposure must be controlled to prevent loss
  - Normal conversation = 40-60 dB
  - Acute damage occurs at 140 dB (jet engine)
  - e.g., operating jackhammer = 100 dB
- Many measures of protection exist:
  - Engineering controls - Reduce noise intensity at source; interrupt noise in path (shielding); reduce reverberation/echo (paneling)
  - Administrative controls – Distance worker from source; reduce noise exposure duration
  - Personal protection – Use aural and semi-aural inserts and circum-aural devices



Recommended permissible exposure

- Noise/sound exposure limits have been determined to prevent cumulative hearing loss:
  - NIOSH says 85 dB exposure should be limited to 8 hrs.
  - Recall sound energy/intensity doubles with every 3 dB
  - i.e., exposure at 88 dB limited to 4 hrs.
  - For every 3 dB increase in exposure, time is cut in half.



Noise dosage and exposure limits

- OSHA sets regulations – Limits are riskier than NIOSH permissible exposure levels (e.g., 16 hrs. at 85 dB).

Duration per day [hours]	Sound level [dBA]
16	85
8	90
6	92
4	95
3	97
2	100
1.5	102
1	105
0.5	110
0.25	115

- Permissible exposure duration can be calculated as:

Noise exposure limit:

L_T [hours] = $\frac{8}{2^{(L_i-90)/5}}$

L_i is the sound level, dBA

Noise dosage = D = $\sum \frac{T_i}{L_T}$

T_i is actual exposure duration
L_T is recommended exposure limit (allowable exposure time)
D > 1 is hazardous exposure level

- Time weighted average (8 hr.) exposure is calculated as:

TWA = 16.61 Log_{10} (D) + 90

Example of noise exposure

- A study found that a worker was exposed to noise at the following levels and times. What is the noise dosage? What is the TWA?

Level [dBA]	Time [hours]
85	3
90	2
92	1
95	2

Duration per day [hours]	Sound level [dBA]
16	85
8	90
6	92
4	95
3	97
2	100
1.5	102
1	105
0.5	110
0.25	115

Noise dosage = D = $\sum \frac{T_i}{L_T}$

D = [(3/16)+(2/8)+(1/6)+(2/4)]

D = 1.104 (not permissible)

Time weighted average (TWA):
TWA = 16.61 Log_{10} (D) + 90

TWA = 16.61 Log_{10} (1.104) + 90

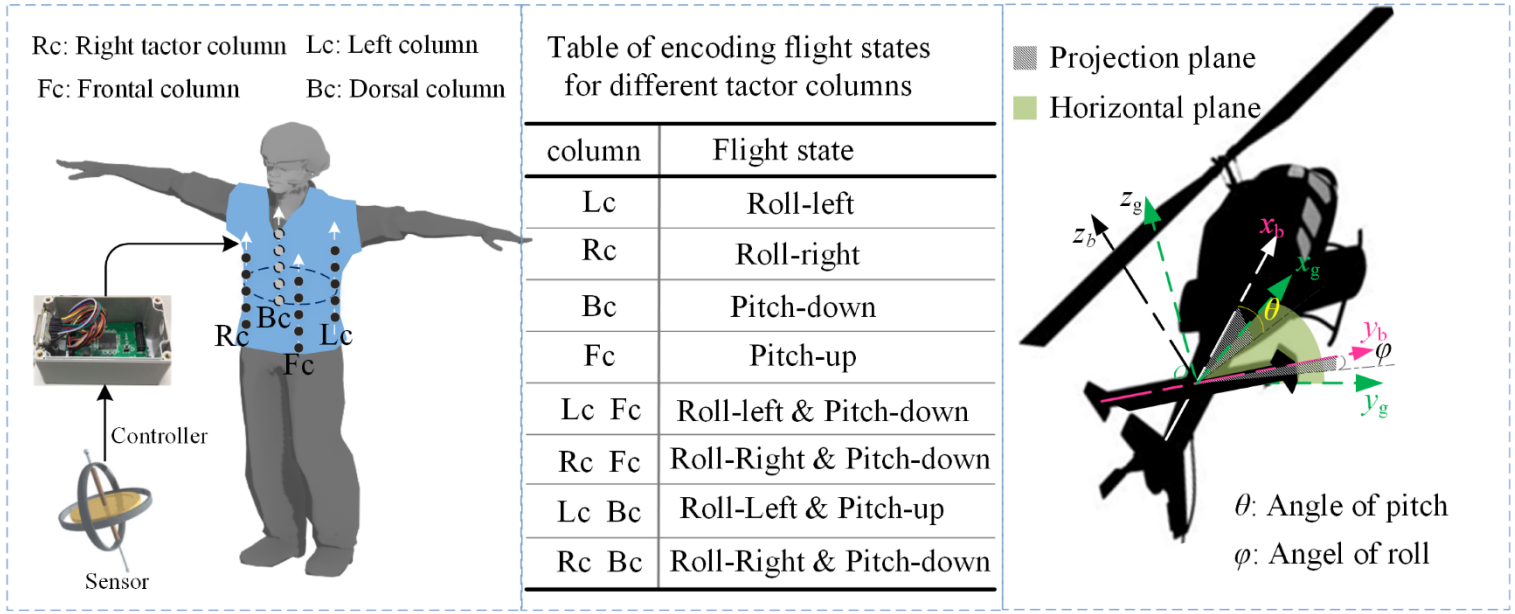
TWA = 90.715 (@ 8 hrs.)

Somatosensory System

- Sensory system that respond to changes at the surface (i.e., skin) or inside of the body
- Can detect: touch, pressure, vibration, temperature, and pain

Human Factors Application: Vibrotactile Displays

- Vision and hearing are often overloaded
- An alternative “channel” for information presentation
- Since hands are often used for controlling devices, displays need to be applied to other, less sensitive, body regions such as the torso



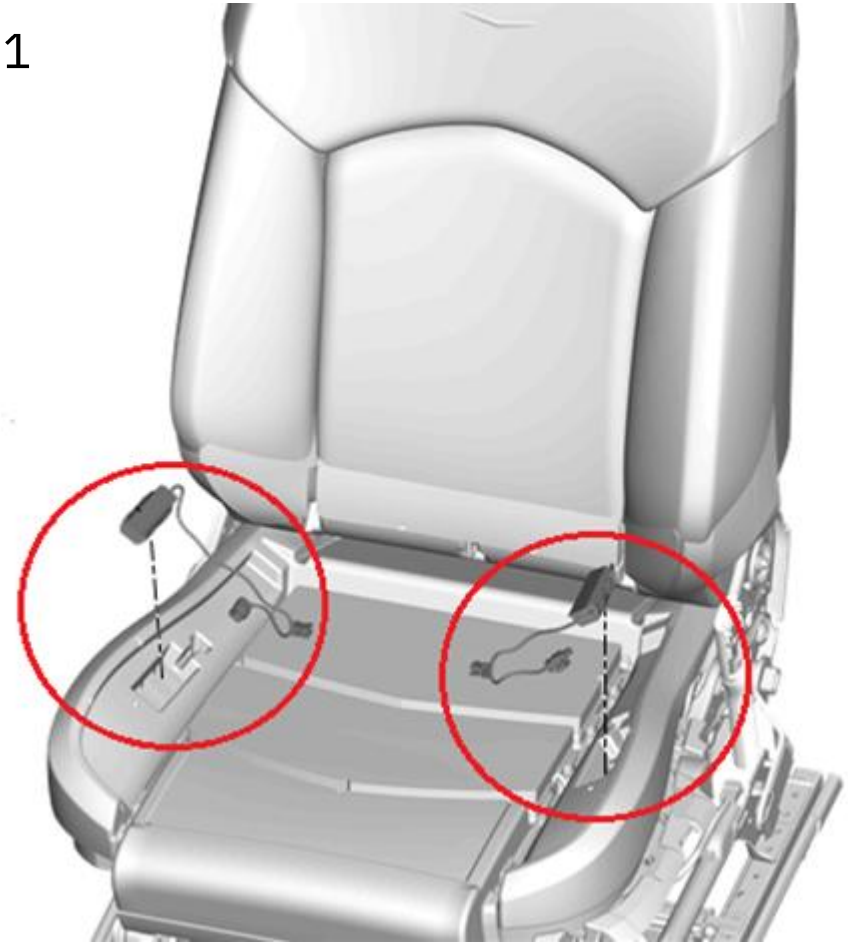
column	Flight state
Lc	Roll-left
Rc	Roll-right
Bc	Pitch-down
Fc	Pitch-up
Lc Fc	Roll-left & Pitch-down
Rc Fc	Roll-Right & Pitch-down
Lc Bc	Roll-Left & Pitch-up
Rc Bc	Roll-Right & Pitch-down

(a) (b) (c)

Ouyang, Q., Wu, J., & Wu, M. (2017). Vibrotactile Display of Flight Attitude with Combination of Multiple Coding Parameters. *Applied Sciences*, 7(12), 1291.

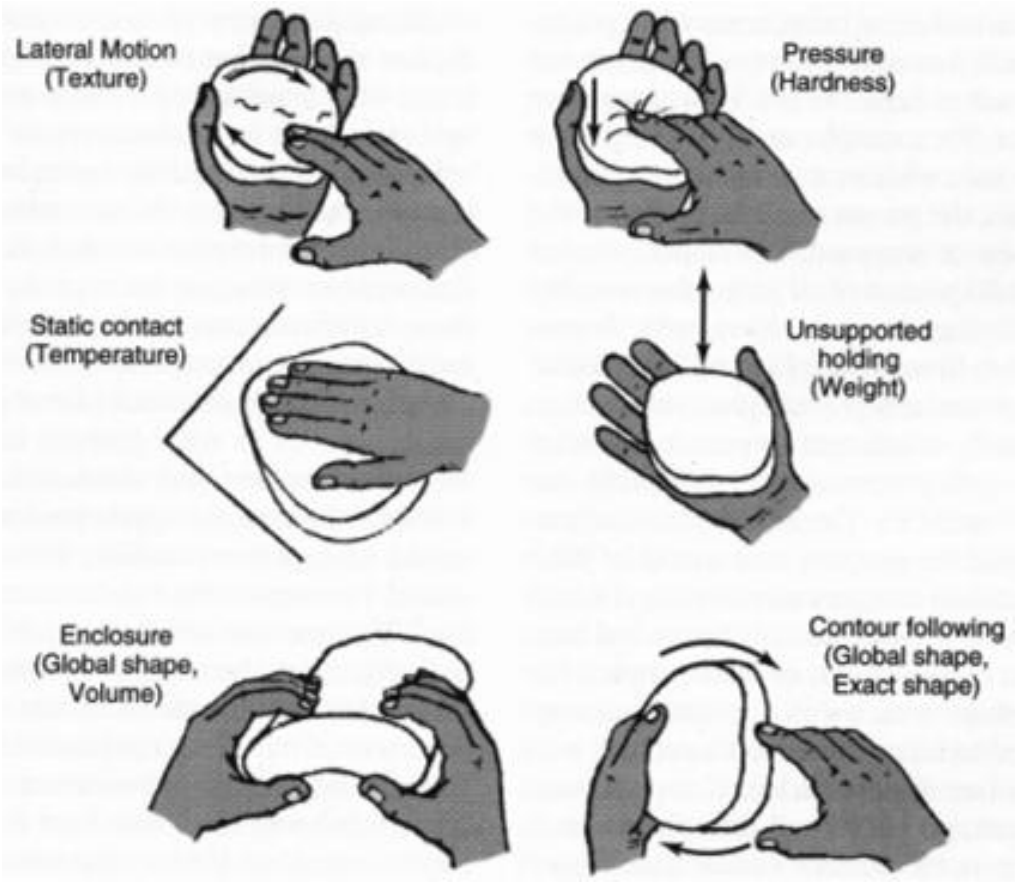
GM Active Safety Seat

https://gm-techlink.com/?p=8091



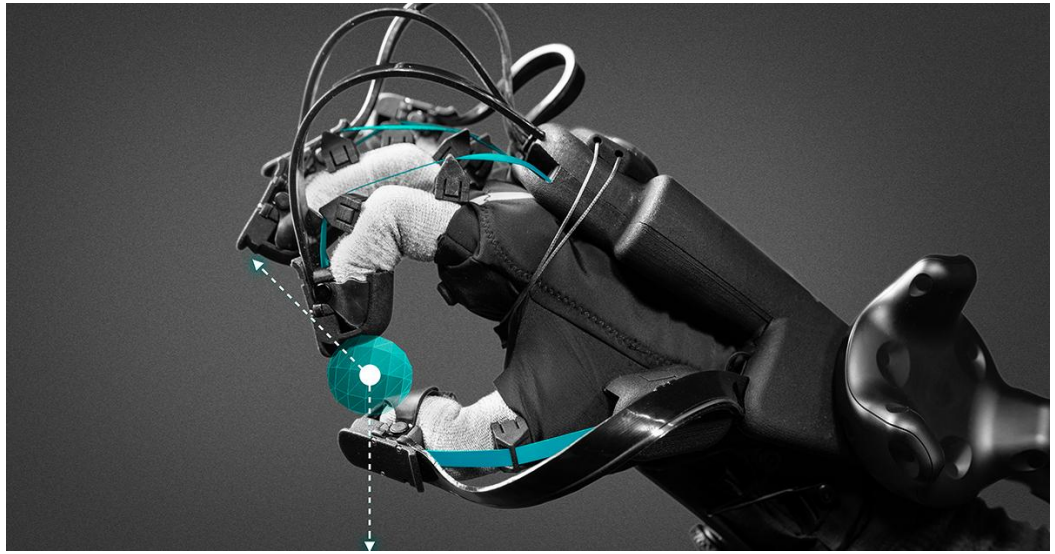
Active Touch (Haptics)

- Perceiver controls the interaction
  - We explore and interact with the world with our hands and fingers
- Our sense of touch allows us to use the various receptors we have in our somatosensory system to understand the world
- Touch acuity is best when fingers move over the object of regard
- Example: landing gear and flaps controls in airplanes (from the first chapter)



Human Factors Application: Haptic Feedback

- Haptic/force feedback is important in contexts like surgery
  - Interactions between the instruments and the patient provides strong haptic cues
- Lack of observable benefits in robot-assisted minimally invasive surgery may be due to lack of haptic feedback
- Challenges exist in terms of:
  - Recording force information at the robot-patient end
  - Presenting force information at the human-interface end



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LEADING THE CHARGE, CHARGING AHEAD